

Empowering Remote Healthcare With Federated Learning for Early Diagnosis of Pulmonary Disease

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Abstract—Recently, the field of healthcare has experienced remarkable technological advancements. However, a considerable challenge persists in providing state-of-the-art healthcare services to individuals in tribal and remote areas. To solve health issues in remote areas, this article introduces a federated learning framework for the early diagnosis of multivariate pulmonary diseases based on cough (voice) samples collected from tribal regions. The proposed framework performs model training on connected local devices, including those living in remote or tribal regions with limited connectivity to centralized servers. The proposed framework ensures the early diagnosis of multivariate lung diseases, such as chronic obstructive pulmonary disease (COPD), to obtain accurate prediction using ensemble learning techniques. The framework applies a convolutional neural network (CNN) to extract discriminatory features from generated spectrograms of cough (voice) samples to classify COPD and non-COPD (healthy) using transfer learning techniques. The proposed framework demonstrates the efficacy of our proposed framework in minimizing resource utilization and model complexity, achieving an impressive accuracy of up to 98.62% with reduced communication rounds and latency, thus facilitating the early diagnosis of COPD. The prototype model of the proposed framework offers an alternative way of a fast diagnosis of respiratory diseases based on cough samples collected from tribal people, which is used with the support of pulmonary and tuberculosis experts (doctors and professionals) for statistical analysis of critical cases of COPD for early diagnosis.

Index Terms—COPD, early diagnosis, federated learning (FL), multivariate pulmonary disease, smart remote healthcare.

I. INTRODUCTION

AS PER findings from the World Health Organization (WHO) [1], chronic obstructive pulmonary disease (COPD) stands as the third leading global cause of human mortality, accounting for 5.4% of the total worldwide deaths. Lower respiratory infections, including conditions, such as

pneumonia and bronchitis, rank among the deadliest infectious diseases globally, representing 4.7% of total deaths in the year 2019–2020 [2], [3], [4]. In addition, a WHO report [1] described a large majority of patients suffering from respiratory diseases around the world. These diseases are categorized into five classes: asthma, lung cancer, interstitial lung disease, COPD, tuberculosis, and lower respiratory tract infection. Examining recorded cough (voice)/lung sound signals through a patient monitoring system is cost-effective due to the fast transmission of recorded samples with less bandwidth requirement of the deployed network [4], [5]. To analyze the discriminatory features extracted from the recorded cough/lung, lightweight models perform fast computation for early diagnosis of subjects (patients) to detect different respiratory diseases automatically. Deep learning (DL) models have gained proliferation due to automatic lung sound analysis for the early detection of multivariate pulmonary diseases.

However, instead of directly accurately predicting multivariate pulmonary diseases from recordings of cough sputum or lung sound signals, current methods have emphasized the detection of respiratory anomalies, i.e., the classification of pulmonary lung sounds (wheeze, stridor, and others). Perna [5] utilized a customized convolutional neural network (CNN) architecture to classify lung respiratory sounds in different classes: 1) chronic disease; 2) nonchronic disease; and 3) healthy diseases using mel frequency cepstral coefficients (MFCCs) features leveraged the highest performance of the model's accuracy of 83%. The multimodal framework is proposed for the early diagnosis of disease using chest radiographs and cough (voice) samples using DL techniques: the proposed work applied a weighted sum-rule fusion method to integrate for accurate prediction of COPD [4], [6]. In [4], the authors used CNN's mix of expert models to predict respiratory diseases in multiple classes. To increase the efficiency of interventions, such as medications, and stop the spread of infectious diseases, it is essential to identify multivariate pulmonary disorders as early as possible.

However, pulmonary experts and professional clinical diagnostic experts deployed spirometry/pulmonary function test (PFT)-based procedures and analyzed computerized tomography (CT) scan images, patient clinical data, and lung sound data to predict and analyze pulmonary health [7]. However, the auscultation of the human lung remains significant to experts and pulmonary professionals. Experts or pulmonary professionals detect different unpremeditated pulmonary sounds during lung auscultations, including stridor, wheezing, and crackling lung respiratory sounds. Deploying a multimodal

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automated framework enabled by federated learning (FL) will benefit tribal communities living in densely forested areas in the early diagnosis of various respiratory diseases. The adoption of FL in remote healthcare settings represents a pivotal advancement in addressing the persistent challenge of timely and accurate diagnosis of critical multivariate pulmonary respiratory diseases and other noncommunicable diseases, COPD, to underserved populations [4], particularly in tribal regions.

FL enables collaborative learning among multiple devices and conducting model training in a decentralized fashion [8], [9]. Furthermore, FL facilitates massive collaboration among healthcare providers and researchers, allowing for aggregating knowledge from diverse multimodal datasets (multivariate pulmonary respiratory data, cough sample data, and other physiological health data) without raw data-sharing [10]. Moreover, to date, such studies do not differentiate between the physiological characteristics that feed the models in predicting low-energy-aware models in tribal regions for the early diagnosis of COPD [11]. The integrated multimodality-based framework assigns equal model weights to selected discriminatory features of various expenses and model priorities. Due to the model selection criteria, the multimodal enabled sensor devices and algorithms bear the associated costs and exponentially increase the complexity and high level of energy consumption of the connected edge devices for early diagnosis of critical pulmonary disease diagnosis.

A. Motivation and Contribution

The primary motivation of this work is to leverage the urgent necessity to enhance healthcare availability in remote and tribal regions. In such areas, people living in dense forest regions depend highly on minor forest produce (honey bees, fruits, and lives) to sustain their lives. They are deprived of the availability of emerging medical facilities for the early diagnosis of critical diseases in tribal communities. Moreover, in densely forested regions, the tribal communities often fail to achieve the needed healthcare solutions and help in accessing healthcare technology and services, resulting in delayed availability of low bandwidth and network connectivity for all the participants (patients, doctors, professionals, and government, and others), which is unrealistic in real-world scenarios. Hence, leveraging the rapid diagnosis of COPD effectively using acquired cough data and resources on heterogeneous participants (edge devices) is still challenging [12]. To address these issues, we investigate how to develop an intelligent patient monitoring framework to ensure early diagnosis of COPD disease in reducing communication delay among heterogeneous participants in FL? The proposed framework employs ensemble learning techniques to extract discriminatory features from acquired cough samples from tribal patients for COPD classification. The significant contributions of this article are addressed as follows.

- 1) This article presents an FL framework for the early diagnosis of multivariate pulmonary diseases based on cough

(voice) samples from tribal regions. By working together and training models using data from different sources, healthcare for less fortunate and distant communities is improved.

- 2) We apply CNN to extract the discriminatory features from generated spectrograms of the cough samples to classify COPD and non-COPD (healthy) using transfer learning techniques.
- 3) We implement a novel framework designed for reliable and energy-scalable ML models, focusing on assessing the impact of statistical features on local devices on both power consumption and execution time for early detection of COPD. Additionally, we demonstrate the efficacy of the proposed multimodal framework in minimizing resource utilization and model complexity, achieving a remarkable accuracy of up to 97.19% with reduced communication rounds and latency, thereby facilitating early diagnosis of COPD.
- 4) The article evaluates multiple DL models, including CNN and long short-term memory (LSTM) with autoencoder, gated recurrent Unit (GRU), and transfer learning models, using the ICBHI 2017 dataset.

B. Article Structure

The structure of this article is as follows. In Section II we offer an extensive review that contributes to the development of the proposed architecture. Section III outlines the methodology employed in this study. The results and comparative analysis can be presented in Section V and Section V-C, respectively. Lastly, in Section VI we summarize our conclusions.

II. LITERATURE WORK

Kumar et al. [4], delved into a new approach for diagnosing COPD through DL techniques. The methodology involved using a CNN to gather, preprocess, and analyze sound recordings from a sample of 70 individuals, comprising 35 COPD patients and 35 individuals without the condition. Beutel et al. [13], focused on analyzing publicly available lung sound data to classify them into seven distinct respiratory diseases, employing a deep neural network. The proposed methodology encompassed three core phases: 1) preprocessing; 2) feature extraction; and 3) classification. Nguyen and Pernkopf [14] introduced a lung sound classification system that utilized an ensemble of CNNs. Multiple CNNs were trained to create this ensemble using various initialization and hyperparameters. Tariq et al. [15] examined the application of a deep neural network featuring four concealed layers to categorize lung diseases. The input data for this neural network consisted of DICOM files. Ma et al. [7] introduced an intelligent digital stethoscope that harnessed a bi-ResNet algorithm to detect lung diseases. This innovative model combined the capabilities of conventional and digital stethoscopes to capture lung sounds, which were subsequently processed by the biResNet algorithm and trained using a comprehensive dataset of respiratory sounds.

III. PROPOSED FRAMEWORK

The proposed framework consists of different steps: 1) collection of datasets; 2) data augmentation; and 3) feature extraction classification, and model training using FL. The detailed description is illustrated in the next section.

A. Dataset Description

We employ public respiratory lung sound databases to perform early diagnosis of respiratory diseases using the FLbased proposed framework.

- 1) *Dataset (DS1)*: We use the ICBHI 2017 [20], a benchmark data repository of lung auscultation sounds database having 920 audio samples from 126 patients presenting various pathological conditions with 5.5 h of audio data. The conditions include a range of states, encompassing individuals in good health and those falling into seven distinct disease categories: Pneumonia, Bronchiectasis (BRON), COPD, URTI, LRTI, Bronchiolitis, and Asthma. Different stethoscopes were employed to record these audio samples (10–90 s with a sampling rate of 4–44.1 kHz with cycles (0.3–15.5 s with approximate 2.6 s) with a total of 6898 respiratory cycles) by leveraging Litt3200, LittC2SE, Meditron, and AKGC417L. We take the 3642 samples as normal cycles; 1864 exhibit cracklings, 886 feature wheezing symptoms, and 506 cycles contain crackling and wheezing.
- 2) *Dataset (DS2)*: RespiratoryDatabase@TR: We use the RespiratoryDatabase@TR [21] lung severity dataset for accurate prediction of respiratory diseases, consisting of 12-channel and four-channel recordings of lung sound signals. Based on recorded data severity, patients (subjects) are categorized into different groups and labeled as COPD0 (five patients), COPD1 (five patients), COPD2 (seven patients), COPD3 (seven patients), or COPD4 (17 patients) with the help of two expert pulmonologists from 41 lung patients.
- 3) *Chest Wall Lung Sound Database (DS3)*: According to [22], the database is one of the most extensive sets of respiratory disease detection lung sounds available to the general public, with 336 lung sound samples by recordings (duration varies from 10 to 50 s) using a Littmann 3200 digital stethoscope at a 4 kHz sampling rate which total dataset is segmented into two classes as healthy or afflicted with one of six conditions: pleural effusion, heart failure, pneumonia, COPD, asthma, or BRON.

B. Data Preprocessing

The collected raw lung sound signals is the proposed framework input, and the preprocessed snipped lung sound signal data represents the output that is represented as $[\kappa] = [\kappa_1[n], \kappa_2[n], \dots, \kappa_T[n]]$, where $\kappa_t[n]$ is denoted as a lung sound signals $[t]$ and $[T]$ denotes the index and number of snippets (discriminatory lung sound data) generated from input

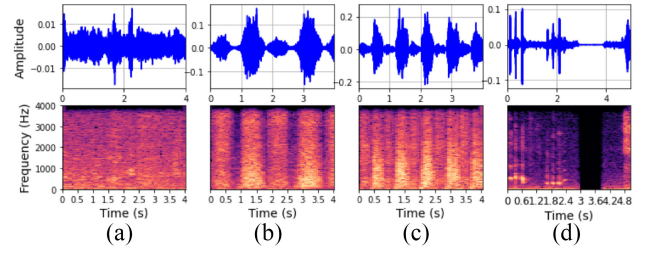


Fig. 1. Spectrogram corresponding to the audio signals. The spectrogram (a) COPD0 (5 patients), (b) COPD1 (5 patients), (c) COPD2 (7 patients), (d) COPD3 (7 patients), or COPD4 (17 patients) with the help of two expert pulmonologists from 41 lung patients.

$L_\alpha[n]$ (shown in (1))

$$\kappa_t[n] = L_\alpha[\eta(t-1)(1-\zeta) + n], t = 1, 2, \dots, T. \quad (1)$$

The collected cough/lung sound signal is complex and nonlinear data consisting of various artifacts and noises while recording lung sound data. Hence, we mitigated baseline wandering (BW) components (0–1 Hz) in the generated sound signals using the discrete Fourier transform (DFT) method. Fig. 1 depicts lung sound signals and their corresponding spectrograms for COPD feature extraction and classification, with the t th DFT temporal lung pulmonary snippet ($\kappa_t[k]$) computed as follows:

$$w_{\text{fm}}[Q] = \sum_{k=0}^{P-1} w_{\text{fm}}[k] \times e^{-j2\pi kQ/P}. \quad (2)$$

C. Data Augmentation

Data augmentation techniques are implemented to address the data imbalance caused by the predominance of COPD cases over minority lung sound classes. The time-stretching (TS) method is used to modify the speed of minority-class lung audio samples while preserving audio pitch. Stretching factors of 0.7 and 0.9 enhanced sample diversity, mitigating imbalance and improving model performance and generalizability across respiratory conditions, even with limited data. We refined the dataset to six categories: pneumonia, BRON, COPD, URTI, bronchiolitis, and healthy, considering random noise, time-shifting (NumPy), pitch modification (constant duration), and TS (e.g., 1.2 or 0.8) for augmentation. Despite residual class imbalances due to COPD prevalence, skewness is reduced by retaining three audio samples per COPD patient shown in Fig. 1 with spectrogram visualizations.

IV. PROPOSED FL-BASED DIAGNOSTIC FRAMEWORK

The diagnostic system's goal is to use the training of global shared models for individual participants (patients) denoted as p_i through the implementation of FL. The proposed approach incorporates FL, which is proven to be a valuable method for using patient medical data. The FL framework functions in five stages, as shown in Fig. 2. The proposed framework motivates the design and development of accurate and robust models, resulting in improved diagnostic precision, treatment effectiveness, and advances in medical research.

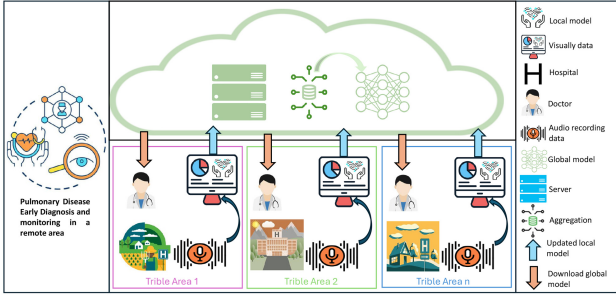


Fig. 2. Proposed FL framework for patient monitoring.

Furthermore, horizontal FL empowers tribal healthcare solutions to collectively address complex challenges that may surpass the capabilities of individual organizations, promoting collaborative efforts to enhance healthcare outcomes on a broader scale. The complete FL process is segmented into different phases as given below.

- 1) *Phase 1 (Initialization)*: In this phase, participating patients transmit their medical data (respiratory cough/lung sound signals) to the edge computing node. The cough/lung sounds are recorded with enabled sensors that are supposed to be trained locally on the wearable device (smartphones and smart watches).
- 2) *Phase 2 (Federated Communication)*: The following initialization, federated communication begins, allowing participant edge models or devices to share knowledge. Each edge device involved in a communication round trains its local model using its dataset. The updates from all patient-enabled edge devices are then sent to a central server, which hosts the global model, as shown in Fig. 2. Federated Averaging (FedAvg) method is used for aggregating the updated model of each patient-enabled edge device. Where ω is the model parameter and $F_k()$ is the local data average gradient in the present model. The aggregations apply in the update model as given by

$$\omega_{t+1} \leftarrow \omega_t - \eta \sum_{k=1}^K \frac{n_k}{n} \nabla F_k(\omega). \quad (3)$$

The weighted sum-rule approach aggregates the collected information to diagnose COPD early. The model receives greater weight from the weighted sum-rulebased fusion procedure, which makes it less errorprone. We compute the mean deviation (MD) value on five-fold (F) cross-validation accuracy for the ModelMFCCs (M1 model) and M2-Fourier Bessel Series expansion (FBSE)+Spectrogram models using the same analogy, as MD directly relates to the consistency of the early diagnostic system. As such, we note that the MD value supports a theoretical study that estimates the technique error rate. We utilize each model's confusion matrix to determine the weights. In Algorithm 1, FL process with a clear separation between the central server and patient-side operations. It defines how the global model is distributed, locally trained on patient devices, and then aggregated back into a global model.

Algorithm 1 COPD-FL Central Server Operation

Input: data sample X, Y , number of federated rounds T, X_i, Y_i, w^{t-1} , number of local epochs E , learning rate η

```

1: initialize  $w^0$ 
2: for  $t = 0, 1, \dots, T - 1$  do
3:   select a set of clients  $S$ 
4:   for  $i$  in  $S$  do
5:     send global model  $w^t$  to patient  $C_i$ 
6:      $\Delta w_i^t, t_i \leftarrow \text{train\_local}(i, w^t)$ 
7:   end for
8:    $\Delta W \leftarrow \sum_{i \in S} \frac{n_i}{n} \Delta w_i^t$ 
9:    $w_t \leftarrow w_{t-1} - \eta \Delta W$ 
10: end for
11:  $\text{train\_local}(i, w^t)$ :
12:    $w_i^t \leftarrow w^t$ 
13:    $t_i \leftarrow 0$ 
14:   for  $e = 1, 2, \dots, E$  do
15:     for every batch  $b$  do
16:        $w_i^t \leftarrow w_i^t - \eta \Delta L(w_i^t, b)$ 
17:        $t_i \leftarrow t_i + 1$ 
18:     end for
19:      $\Delta w_i^t \leftarrow w^t - w_i^t$ 
20:   end for
21: return  $\Delta w_i^t, t_i = 0$ 

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- 3) *Phase 3 (Local Model Training)*: In a hospital environment, the initiation of local model training involves the central server disseminating the initial global model parameters to each hospital. Hospitals then conduct training sessions on their local models using Stochastic Gradient Descent (SGD) with their specific datasets. Local model parameters are updated through gradient computations based on the loss function and regional data.
- 4) *Phase 4 (Model Update)*: After completing each iteration of local model training, hospitals transmit their adjusted model parameters to the central server. Subsequently, the central server enhances the global model by computing the average parameter updates received from all participating hospitals.
- 5) *Phase 5 (Model Aggregation)*: The proposed framework performs the FedAvg at local and global models (see Algorithm 2), which stands out from traditional distributed optimization methods by aggregating only model weights, not raw data, from each hospital. The weights represent what local models have learned from their respective hospital datasets. FedAvg is designed for scalability, making it suitable for large-scale FL scenarios involving numerous hospitals.
- 6) *Phase 6 (Global Update)*: Upon gathering the revised model parameters from all hospitals, the central server dispatches fresh global model parameters to each hospital, commencing the subsequent round of local model training. This iterative sequence of model adjustment,

Algorithm 2 Local and Global Federated Aggregation**Input:** Number of rounds N

```

1: Initialize  $W^0$ 
2: for  $P = 1, 2, \dots, N$  do
3:   Select a set of models/devices  $Q$ 
4:   for  $k = 1, 2, \dots, Q$  do
5:      $W_{D''k}(P) \leftarrow W_{D''k}(P-1) - \eta \delta L_k(W_{D''k}(P-1))$ 
6:   end for
7:   Local Model Updates after  $k_1$ :
8:   for  $D'' = 1, 2, \dots, d$  do
9:      $W_{D''}''(P) \leftarrow \text{Aggregation at local devices } (W_{D''k}(P))$ 
10:  end for
11:  After  $k_2$  Global Update:
12:   $W(P) \leftarrow \text{GLOBALAGGREGATION } (W_{D''}(P))$ 
13: end for
14: LocalAggregation ( $D'', W_{D''k}(P)$ ):
15:
16: return  $W_{D''}(P) \leftarrow \sum_{k \in \mathcal{D}_{D''}} \frac{D_{D''}}{D} W_{D''k}(P)$ 
17: GlobalAggregation ( $W_{D''}(P)$ ):
18:
19: return  $W(P) \leftarrow \sum_{D'' \in \mathcal{d}} \frac{D_{D''}}{D} W_{D''}(P) = 0$ 

```

consolidation, and global enhancement persists until the global model reaches the target accuracy level.

A. Heterogeneous Network and Scalability

The proposed framework deals with heterogeneous interacted participants (p_i) and collected heterogeneous lung sound signals/cough sample data captured from connected edge devices. To reduce data preprocessing time and improve the scalability of local models for early diagnosis of COPD, we divided the connected participants (p_i) into different clusters. We investigate the following problem: “How can we obtain high-level performance while training local models on heterogeneous participants (p_i) in FL?” To solve this problem, we used the clustering-aware method to divide heterogeneous participants.

1) *Clustering of Participants (p_i):* Clustering is essential for grouping participants into clusters based on commonalities, such as closeness to one another geographically or similarity in health problems. For several reasons, this clustering process is advantageous. The first benefit is that localized cooperation is possible within each cluster, reducing communication latency and encouraging effective data interchange between participants. Effective and top-tier training of the globally shared model M among all participants faces challenges due to differences in bandwidth availability B_i , dataset size $|D_i|$, and participant-specific factor F_i , where $1 \leq i \leq N$. These parameters show different values among the participants, causing variations in training time and quality. Some participants perform better, while others experience significant delays and provide inadequate training. Combining the weight-parameter matrices from these heterogeneous contributors produces a less-than-ideal merged matrix and demands a more extended training duration. Consequently, the performance of participants is severely compromised. To overcome this challenge,

the N participants are divided into k clusters, denoted as $[C_1, \dots, C_k]$. Each cluster $[C_j]$ consists of $[N_j]$ participants, where $1 \leq j \leq k$ (where $j = P$), and the sum of N_j in all clusters equals N .

2) *Clustering Criteria:* This study incorporates three parameters, denoted as B_i , $|D_i|$, and F_i , where $[1 \leq i \leq N]$, to examine their contributions. Here, we introduce three variables λ_1 , λ_2 , and λ_3 , show the weights allocated to $[B_i, |D_i|]$, and F_i , respectively, due to the varying significance of each parameter. Additionally, we evaluate the strength $S(p_i)$ of participant p_i using the following estimation:

$$S(p_i) = \alpha \lambda_1 B_i + \lambda_2 |D_i| + \beta \lambda_3 F_i \quad (4)$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$. The specific scenario determines the values of λ_1 to λ_3 . If bandwidth B_i is more important, λ_1 is greater than λ_2 and λ_3 . Similarly, if the dataset size $|D_i|$ at a participant holds greater significance, λ_2 is greater than λ_1 and λ_3 . To ensure comparability and dimensionlessness with $|D_i|$, the constants α and β are used to normalize B_i and F_i , respectively. As a result, $S(p_i)$ becomes a dimensionless quantity, enabling meaningful comparisons and analysis. Based on the participant's strengths $S(\bullet)$, we have divided N participants into k clusters. This intends to weaken each cluster individually while enlarging the space between them. All participants who declare a desire to join the federation send queries to the server, which then begins the clustering process. After collecting data from each participant, the server determines the strength $S(\bullet)$ for each participant and creates k clusters.

3) *Generation of Models at Server:* FL has been explained in Algorithm 1. The proposed framework is evaluated by analyzing its performance in various models, aiming to identify appropriate local and global models for integration.

4) *Training of Local Models:* The mathematical foundation of computed feature vectors using these gates update gate parameters for the lung sound signal time (t) is given as follows:

$$z_t = \sigma(W^z x_t + U^z h_{t-1}). \quad (5)$$

The symbols W^z and U^z represent the weights for the current and previous reset gates in a GRU model applied to lung sound signals, with the activation function $[\sigma]$ mapping outputs between 0 and 1, and the reset gate parameters at time (t) are computed as specified in

$$r_t = \sigma(w^r x_t + U^r h_{t-1}). \quad (6)$$

The reset gate is initially employed to generate a current memory unit using the Hadamard product ($\odot$) based on distance measure method with tanh activation function $[-1, 1]$ to solve the issue of explosion gradient h'_t . We calculated the last memory unit at the current time (t) is performed

$$h_t = (z_t \odot h_{t-1} + (1 - z_t) \odot h'_t). \quad (7)$$

In contrast to CNN auto-encoders, which concentrate on spatial features, the GRU model effectively retains pertinent historical information through tunable gates z_t and $1 - z_t$, providing computational efficiency and low memory demands.

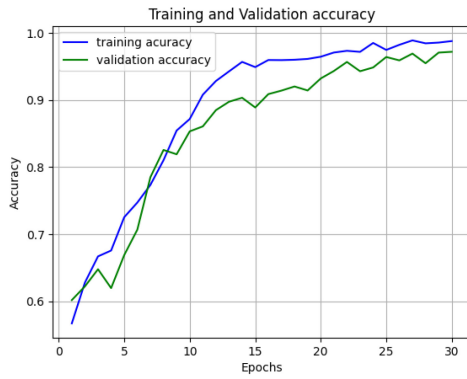


Fig. 3. Classification accuracy.

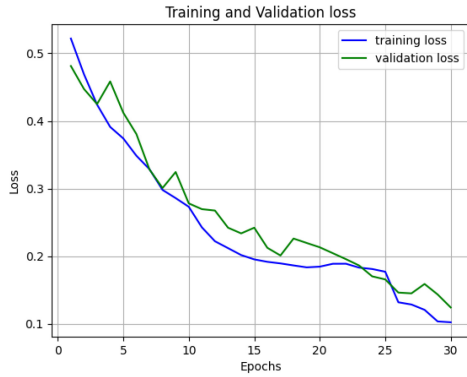


Fig. 4. Classification loss.

It is superior at capturing temporal dependencies in respiratory data, making it the perfect choice for diagnosing respiratory disease features $[\max_{D_i, D'_i} \|f(D_i) - f(D'_i)\|_1]$ as given in [20].

$$\Delta p = \|f(D_i) - f(D'_i)\|_1. \quad (8)$$

B. Experiment Setup

The proposed solution is implemented using Internet of Medical Things (IoMT) enabled edge devices connected to a central server. Each patient's edge device is equipped with a dataset containing 237 unique lung sound audio samples from COPD patients. This dataset is split in an 80%:20% ratio for training and validation testing, following a five-fold cross-validation approach. The framework utilizes the FedAvg method available in the Flower framework [13] to aggregate updates from all edge devices within the federated communication network.

V. RESULT AND DISCUSSIONS

We conduct a comprehensive analysis to evaluate the effectiveness of the FL framework, encompassing various models, including the CNN model featuring an autoencoder, LSTM, GRU, EfficientNet-B0, MobileNet-V2, VGG-19, and the FL model with GRU. The study aimed to assess and compare these models in terms of their diagnostic accuracy and efficiency for respiratory diseases. As presented in Table I, the proposed framework leverages the CNN model incorporating an autoencoder method that achieved an accuracy

TABLE I
AVERAGE (%) ACCURACY

Models	V. Accuracy	T. Accuracy
LSTM	85.64 %	86.44 %
EfficientNet-B0	86.36 %	85.28 %
GRU	88.47 %	90.73 %
CNN (autoencoder)	89.07 %	87.49 %
MobileNet-V2	91.34 %	88.31 %
VGG-19	96.84 %	95.53 %
Proposed FL (GRU)	97.19 %	97.03 %

Abbreviation: V= Validation, T= Test

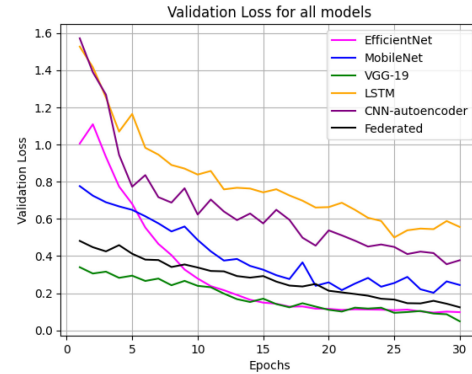


Fig. 5. Validation loss.

rate of 89.07%. In contrast, the LSTM and GRU models yielded accuracies of 85.64% and 88.47%, respectively. Notably, the MobileNet-V2 model exhibited an impressive accuracy of 91.34%, surpassing the EfficientNet-B0 model, which achieved an accuracy of 86.36%. Among the models employed, VGG-19 stood out with an exceptional accuracy of 96.84%. However, the FL model utilizing GRU demonstrated remarkable accuracy at 97.19%, establishing itself as the most proficient model among the choices. These findings underscore the higher accuracy of the FL model, surpassing the performance of other existing methods.

In Figs. 3 and 4, we observe the graphs representing the training validation loss versus epochs and validation accuracy of the proposed framework as these graphs show the accuracy varies with the number of epochs during the classification process. Furthermore, Figs. 5–7 show validation loss and accuracy for all the models that have been put into practice and the number of epochs. The FL's classification performance is shown in Table II, where recall gauges the model's capacity to detect every positive instance and precision assesses the accuracy of positive predictions. The proposed framework diagnoses COPD with a weighted average accuracy of 97% and a macro average accuracy of 96%. Fig. 8 shows the confusion matrix corresponding to the proposed model for six respiratory disease classes. It showcases different permutations of class results, encompassing correct positives, correct negatives, incorrect positives, and incorrect negatives.

A. Analysis of Model Latency and Power Consumption

Figs. 9 and 10 shows latency the end-to-end power consumption of edge devices relative to total latency time and the number of communication rounds in the FL environment,

TABLE II
ACCURACY (%) FOR EARLY DIAGNOSIS

Class	Precision	Recall	F1-score	Support
COPD	0.95	1.00	0.97	15
Bronchiolitis	0.94	0.94	0.94	14
Bronchiectasis	1.00	0.95	0.98	48
Pneumonia	0.88	1.00	0.94	22
URTI	0.97	1.00	0.99	31
Healthy	1.00	0.89	0.95	22
Accuracy	NA	NA	0.98	150

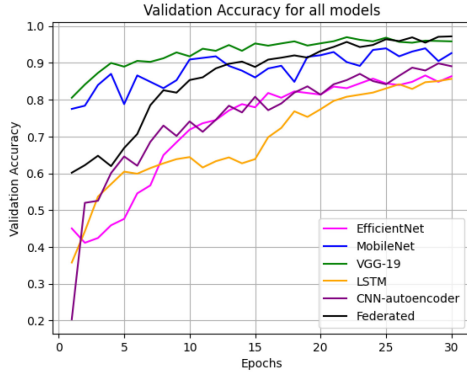


Fig. 6. Validation accuracy.



Fig. 7. Prediction time.

further emphasizing the advantages of our approach in terms of both speed and energy efficiency. Our proposed method demonstrates significantly lower energy consumption (measured in Joules per unit time) compared to existing approaches.

B. Scalability and Heterogeneous Analysis

We analyze the model performance based on each patient in patient clusters. N patients are divided into k clusters as $C_1, C_2, \dots, C_k$. Each cluster C_j consists of N_j patients where $1 \leq j \leq k$, and the sum of N_j across all clusters equals N . We examine two options to assess scalability and heterogeneity: 1) keeping the final weights of the leaving patient for subsequent aggregations and 2) disregarding prior weights when a patient departs and just aggregating new patient weights. The performance analysis, shown in Table III, highlights results when each of the initial five patients is removed and a new patient is added. In most cases (4 out of 5 for loss, 5 out of 5 for accuracy), better results are achieved by excluding the final weights of the departing patient.

TABLE III
ADDING AND REMOVING PATIENT

Patient	Option-1		Option-2	
	Loss(%)	Accuracy(%)	Loss(%)	Accuracy(%)
Patient1	3.269	0.657	3.748	0.589
Patient2	3.676	0.754	4.644	0.682
Patient 3	3.146	0.754	4.261	0.722
Patient4	3.93	0.756	4.899	0.705
Patient5	3.395	0.759	3.785	0.726

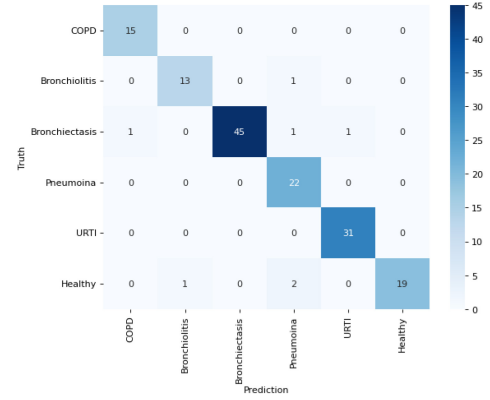


Fig. 8. Confusion matrix.

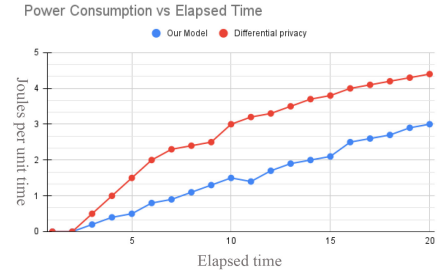


Fig. 9. End-to-end power consumption versus elapsed time.

C. Comparative Analysis With Existing Methods

In this article, the approach utilized demonstrates superior performance compared to existing methodologies. By employing an FL framework with GRU as the local and global models, our approach achieves better results compared to various established models, including CNN, GRU with Leaky ReLU activation, Snapshot Ensemble of CNN models, 2D-CNN with Normalized EBU Value Augmentation, Bi-ResNet DL Model, transfer learning using VGG-19, and the boosted decision tree algorithm. The performances of the models implemented in the existing approaches are outlined in Table IV.

D. Complexity Analysis

The proposed framework shows that the boosted decision tree algorithm takes $O(ZN^2)$ time complexity to constitute an ensemble decision tree of cough voice/lung sound signal features, where Z and N show some selected discriminatory features. The constructed boosted decision tree models need to update the selected features $O(h \times Z)$ time as early as the new fused feature instance arrives; $[h]$ is the height of the decision tree. Furthermore, the time complexity of the

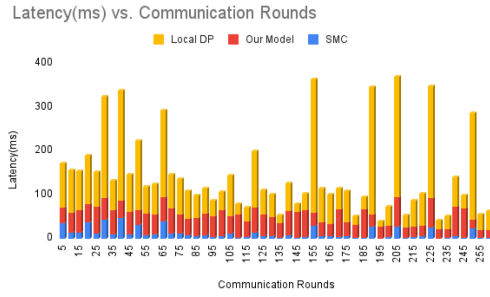


Fig. 10. Upload latency (ms) versus Communication round.

TABLE IV
ACCURACY COMPARISON OF EXISTING METHODS

Methodology	Accuracy (%)
CNN (MFCC) [4]	89.9
GRU (leaky ReLu) [13]	95.67
Ensemble of CNNs [14]	91.4
2D CNN (EBU normalization) [15]	97
Prop. FL(GRU)	97.19

proposed framework using CNNs ensemble, 2D-CNNs and FL models as $O(\sum_{s=0}^D P_{s-1} \cdot F_s^2 \cdot f_s \cdot X_s^2)$, D , and P show layers and the input multimodal. The defined terms F and f show the spatial filter with kernel size $[3 \times 3]$ and several mask filters. $[X]$ depicts the total size of the output feature map obtained in the s th layer. We leverage the working of ensemble-based learning methods and CNN models (2D-CNN, CNN ensemble) models as COPD diagnostic models for accurate diagnosis of the execution of multivariate pulmonary diseases in parallel. Therefore, the total time complexity of the implemented COPD diagnosis algorithms $O(P \times F^2 \cdot f \cdot X^2)$ as measured time complexity of customized 2D-CNN and an ensemble of CNNs in the FL settings for COPD diagnosis is higher time complexity as compared to the ensemble learning techniques.

E. Performance Analysis in Multimodal Data Fusion

The proposed framework extracts MFCCs and FourierBessel Series Expansion (FBSE) features from DS1, DS2, and DS3 lung sound signal datasets. The MFCCs are used for audio signal analysis as they mimic the human auditory perception of sound to distinguish subtle variations in lung sounds or cough signals caused by respiratory abnormalities. Then, FBSE provides Bessel function sets to decompose lung sound signals into orthogonal components, enhancing feature extraction, enabling detailed energy distribution analysis, and detecting subtle, nonlinear patterns in nonstationary lung sound signal classification. Moreover, we use spectrograms to visualize the time-frequency evolution of sounds, which is crucial for the analysis of transient and periodic signals. We compute the MD value on five-fold (F) cross-validation accuracy for the model-MFCCs (M1) and FBSE+Spectrogram (M2) using the same analogy, as MD directly relates to the consistency of the early diagnostic system. As such, we note that the MD value supports a theoretical study that estimates the technique error rate. We utilize each model's confusion matrix to determine

TABLE V
ACCURACY ON DIFFERENT FEATURES AND WEIGHTS

Modality	Weight	Mean accuracy (S_i)	W_{sm} score
M1-MFCCs	0.31	91.4	0.28
M2-FBSE	0.30	89.2	0.26
M1-Spectrogram	0.39	93.2	0.36

TABLE VI
PERFORMANCE METRICS

Task	Data	Metric	F 1	F 2	F 3	F 4	F 5	Avg
T1	DS1,	Accuracy	93.5	95.9	95.48	98.9	97.9	96.6
	DS2,	Precision	97.7	98.9	96.6	96.9	97.8	97.7
	DS3	Recall	96.45	96.79	96.92	97.71	96.98	97.94
T2	DS1	Accuracy	98.99	99.95	99.96	99.86	99.94	99.96
		Precision	99.98	98.96	98.99	99.96.0	99.9	99.89
		Recall	98.95	99.96	98.86	98.89	98.75	99.96
T3	DS1,	Accuracy	93.99	95.76	94.89	95.98	94.93	94.97
	DS2	Precision	92.93	91.93	92.96	91.93	92.95	93.95
		Recall	92.67	92.96	93.94	91.94	91.50	92.58

the weights (see Tables V)

$$MD = \frac{1}{n} \sum_{i=1}^n |x_i - S|. \quad (9)$$

The S_1 and S_2 similarity are given as follow:

$$S_1 = \sum_{i=1}^5 \left(\frac{\text{Model-1}_{MFCCs} - \text{accuracy}(A_i)}{5} \right) \quad (10)$$

$$S_2 = \sum_{i=1}^5 \left(\frac{\text{Model-2}(A_i)}{5} \right). \quad (11)$$

The weight calculations ($W_2 = 0.46$ and $W_1 = 0.56$) for each model (M1 and M2). The accuracy computes using the weighted sum-rule fusion method is 98.62%. FL framework combines extracted mel-spectrograms, MFCCs, and FBSE features for analysis of severity of COPD diseases based on open datasets (shown in Table V). Three tasks are used in the study to classify respiratory diseases using FL techniques: T1 uses lung sound samples from combined datasets (DS1, DS2, DS3) to classify seven diseases; T2 uses a fusion method to classify six diseases using only DS1 data; and T3 uses DS1 and DS3 to target overlapping symptoms of asthma and COPD. Cross-validation and feature selection assess the framework's performance on these tasks. The findings are shown in a comparison Table VI and show how well the system handles a variety of datasets and illness complexity while maintaining data decentralization.

F. Analysis and Model Interpretability

FL framework extracts FBSE, MFCC, and spectrogram-based features from the cough/lung sound signals for early diagnosis and accurate prediction of COPD. The most common parameters of traditional diagnostic methods for differentiating asthma and COPD are Forced expiratory volume (FEV) in one second divided by Forced Vital Capacity (FVC) ($FEV1/FVC \leq 0.70$) for COPD patient and $FEV1/FVC \geq 0.70$ for asthma patient, with a percentage change of FEV1 value as per baseline after taking bronchodilators ($FEV1 \geq 12\%$

for COPD and $FEV1 \leq 12\%$ for asthma). PFT/spirometry effectively distinguishes asthma from COPD using GINA and GOLD protocols when the patient practices this problem to perform diagnostic tests. The comparative analysis of overlapped symptoms (COPD, Asthma, and other respiratory diseases) through defined tasks (T1, T2, and T3) are shown in Table VI.

VI. CONCLUSION

In this article, we introduced a novel framework for the early diagnosis of respiratory diseases using FL techniques. The proposed framework classifies several diseases, such as asthma, bronchiolitis, BRON, COPD, and healthy, LRTI, pneumonia, and URTI. FL presented a pragmatic solution for decentralized model training of healthcare services. In particular, the investigation revealed that the FL paradigm surpasses all alternative models scrutinized, attaining remarkable training and validation accuracy levels, reaching 98.79% and 97.19%, respectively. The results demonstrated the efficacy of FL framework in minimizing resource utilization and model complexity with reduced communication rounds and latency, thus facilitating the early diagnosis of COPD. The impressive results substantiate the model's capacity for robust generalization. In the future, we will implement the optimized working prototype model leveraged with lower power and cost effectiveness for the fast diagnosis of COPD patients based on recorded cough audio samples in tribal or remote areas.

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